tec-speed 30.0 VT-370X / VT-370X LCTE

UL Approval: E214381
Version:16/10/2025

PROCESS GUIDE

Precautions in Handling

Storage Condition & Shelf Life

		Laminate
Storage Condition	Temperature	Room
	Relative Humidity	/
Shelf Life		24 Months(airproof)

- Laminates should be stored flat in a clean dry area. Cores will benefit from being stored between two stiffeners in order to prevent unnecessary bending of the layers or damaged corners. Soft slip sheets should be used to prevent dust and debris from be embossed into the material.
- Laminates are without woven glass reinforcement and are softer than most other rigid base materials for printed circuit board and are more susceptible to handling damage. As with all laminate materials, especially thin materials, care should be taken to avoid distorting the material during handling.

Designing and Inner layer Process

- Materials are compatible with many tooling systems. Choosing whether to use round or slotted pins, external or internal pinning. Whichever approach is used, it is good practice to retain copper around tooling holes.
- A chemical process consisting of organic cleaners and a micro-etch is the preferred method of preparing copper surfaces for coating with liquid or film photoresist. Mechanical scrubbing is not recommended.
- Developers, strippers, and copper etchants used to process epoxy glass materials will also work with layers. Thin cores may require leader boards for conveyorized processing and frames or supportive racks for vertical-type processing.
- Laminates are compatible with most oxide and oxide alternative processes. Highly caustic, high temperature processes, should be followed by a thorough rinse and bake of the inner layers.

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Inner Layer Artwork Compensation Factors

Thickness	Configuration	Direction	Comp.(ppm)
5mil 1/1oz	Signal/signal	Length / Width	900 / 500
5mil 1/1oz	Signal/Ground	Length / Width	850 / 450
5mil 1/1oz	Ground/Ground	Length / Width	600 / 300

If you use other thickness types, please contact Ventec technical service engineer.

Laminates Availability

Laminate	Core Thickness (inch)	Dk@10GHz	Df@10GHz
		IPC-TM650 2.5.5.5 Clamped Stripline	IPC-TM650 2.5.5.5 Clamped Stripline
VT-370X	0.0035	2.90	0.00035
	0.0040	2.90	0.00035
	0.0045	2.90	0.00035
	0.0050	2.90	0.00035
	0.0080	2.90	0.00035
	0.0100	2.90	0.00035
VT-370X LCTE	0.0035	2.95	0.00055
	0.0040	2.95	0.00055
	0.0045	2.95	0.00055
	0.0050	2.95	0.00055
	0.0080	2.95	0.00055
	0.0100	2.95	0.00055

Copper Type	Copper thickness	Remark
HVLP3	Hoz, 1oz	Standard
HTE	Hoz, 1oz, 2oz	Optional
RTF	Hoz, 1oz, 2oz	Optional
RA	Hoz, 1oz	Optional

- Copper foil: Hoz~2oz, HTE, RTF, RA and HVLP are all available based upon request.
- More types may be available upon request.

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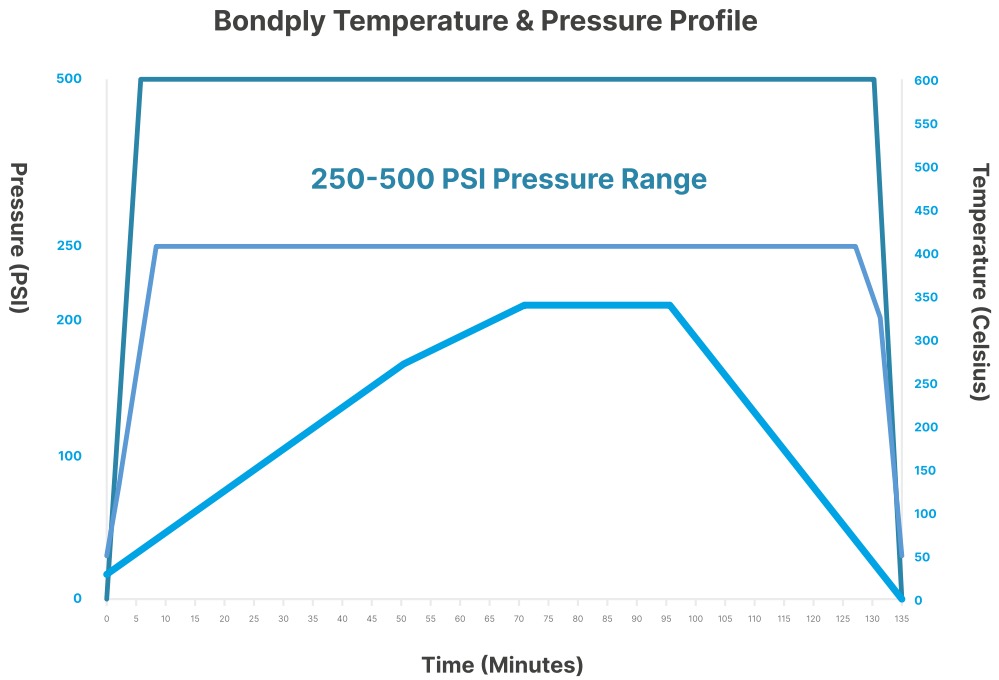
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Bonding for Multilayer

- Special pretreatments of etched surfaces using plasma processes shouldn't be necessary providing care was taken to protect the substrate surface after copper etch. Inner-layers should be baked at 120°C for 60 minutes to ensure removal of volatile substances prior to MLB bonding.
- Cores are compatible with a broad range of thermosetting adhesive systems. Many factors, such as electrical performance, flow characteristics, ease of processing, and bond temperature requirements are considered when making the best overall choice.
- The press cycle is determined by the requirements of the chosen adhesive resin system of High Tg prepreg.

Bond cycle

1.Heating rate (Rate of Rise) of material [Material Temperature]: 50-315 °C: ≥5.5°C/min 315-371 °C≥2.7°C/min
2.Curing Temperature & Time: 30-60min at more than 371°C.
3.Full Pressure: 250-500psi should be applied full pressure before 23 °C.
4.Cooling rate of material: 260-121 °C ≤1.1°C/min.
5.Cushion for pressure evenness is needed.



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Mechanical drill parameters

Drill Bit Diameter (mm)	Spindle (KRPM)	Infeed (IPM)	Retract (IPM)	Hit Counts
0.3	150	47	300	300
0.5	110	50	400	300
0.8	80	55	400	300
1.0	55	60	400	300

- Excessive wear of carbide drill bits. New drill bits are recommended.
- Rigid entry and exit material is critical in order to remove any debris or deposits from the drill bit. 10~25mil (0.25 –0.6 mm) phenolic entry is acceptable for most applications.

Desmear Process

Plasma treatment is necessary process, and material should be baked more than 1 hour at 120°C before plasma treatment. Meanwhile, material cannot be suffered any high pressure washing and scrubbing before metallization.

Plasma: Half time of FR-4 conditions is recommended.

Pressure (mT)	Temperature (°C)	Gas mixture	Power (W)	Duration (min)
220	60	3.50 l/min O_2	7000	10
220	60	0.19 l/min CF_4 , 3.50 l/min O_2	6800	10
220	60	0.30 l/min N_2	6800	20
220	60	1.0 l/min H_2 , 1.0 l/min N_2	7000	10

Permanganate Desmear: Not recommended.

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Plating and EDS

- Laminates have been engineered to be highly resistant to chemical infiltration and a robust hole wall preparation process is necessary for a successful deposition plating process. Following hole wall preparation, they will accept standard electroless copper or direct deposit metallization plating process.
- **Image, Develop, Etch, Strip Film.**
  When copper surface preparation is required, chemical cleaning processes are preferred. Mechanical scrubbing should be avoided due to possible mechanical damage. Otherwise, standard processing should be used.

Solder mask and finish

- Materials should be rinsed and baked prior to solder mask application. Rinsing in warm or hot water for 20-30 minutes followed by 60 minutes @125°Cshould be sufficient, especially if the bake is done under vacuum.
- No special treatment is required if the surface has not been mechanically scrubbed. In rare cases, where adhesion is poor, a plasma treatment may be used to activate the exposed PTFE.
- Laminate is compatible to most final finishes (HASL, Sn, Ag, Ni/Au, OSP, etc). A rinse/bake should be done prior to HASL. Make sure etched substrate surfaces are clean before ENIG plating, or else background plating could occur.

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Routing

Drill Bit Diameter (mm)	Spindle (KRPM)	Infeed (IPM)	Retract (IPM)
0.8	50	10	500
1.0	40	12	500
1.2	34	12	500
1.6	25	18	700
2.0	20	27.2	700
2.4	20	35	700
3.0	20	42	700
3.2	20	42	700

- Excessive wear of carbide drill bits and Diamond coated drill bits are recommended.
- Rigid phenolic entry and a rigid backer should be used. In some cases, adding paper (white paper or Kraft paper) between the phenolic and the part allows better conformance to surface topography and may reduce burring.
- Materials will generally provide a better edge quality due to none glass fabric.

Packaging and baking recommendation

- It is recommended to bake the board before packaging at 125°C/4~8h to avoid moisture causing a decrease in heat resistance.
- If the PCBs needs to be stored for a long time before use, it is recommended to use aluminum foil vacuum packaging.
- If exceed 3 months after packaging, It is best to bake the PCBs at 125°C/4~6h before assembly before use.